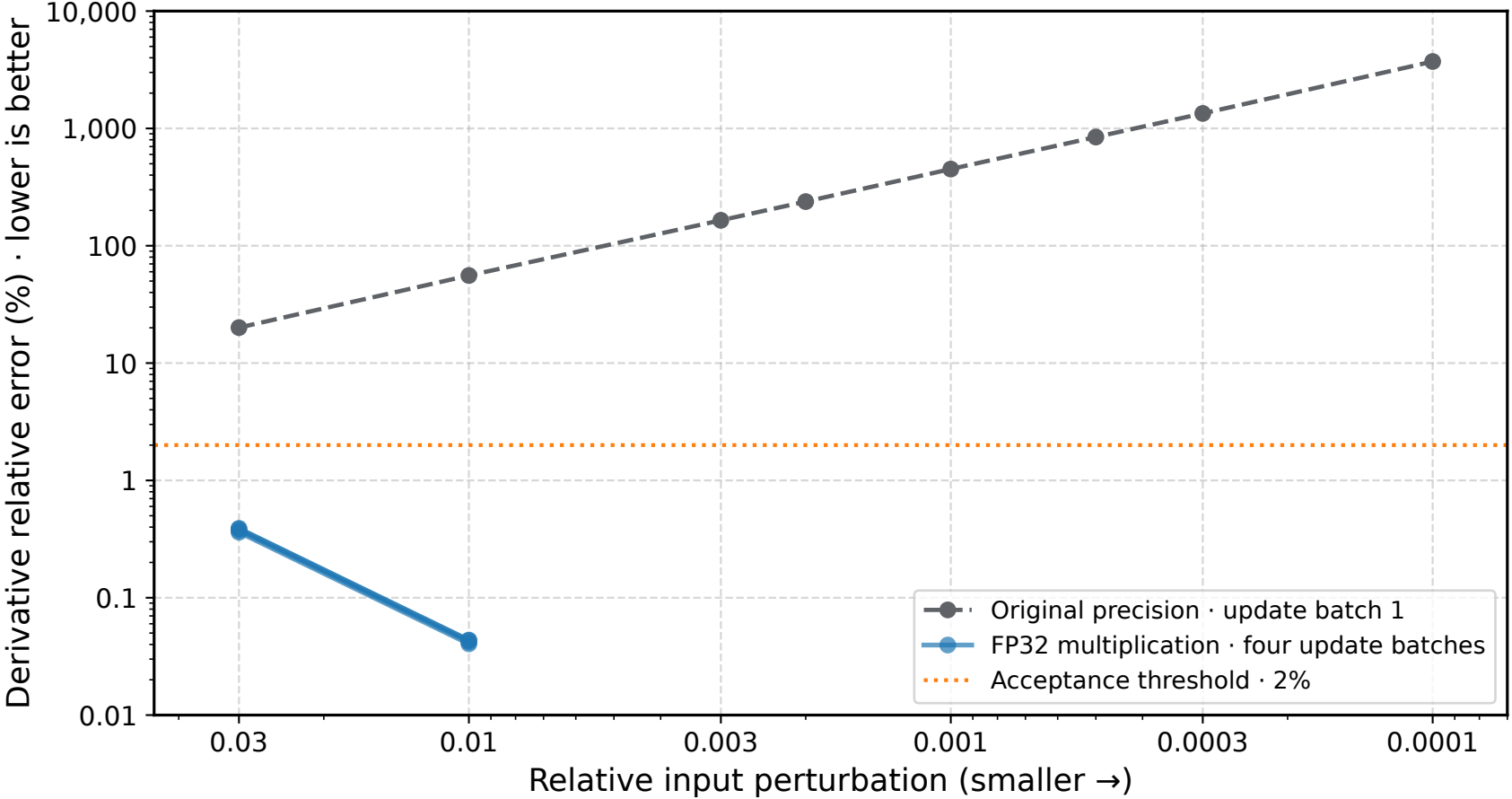
Derivative relative error (%) · lower is better
10,000
1,000
100
10
1
0.1
0.01
0.03
0.01
0.003
0.001
0.0003
0.0001
Relative input perturbation (smaller →)
Original precision · update batch 1
FP32 multiplication · four update batches
Acceptance threshold · 2%